

**Global continuous phenomics reveals trait- and scale-specific
environmental geography of thistle capitula**

Blinded submission candidate — scientific HOLD retained

Abstract

Aim

Broad-scale plant trait datasets often summarize taxa by species means or coarse categorical states. We asked whether public photographs can instead recover repeated continuous capitulum phenotypes and determine how the present geography of each trait aligns with predeclared environmental gradients at head, photograph, within-taxon, among-taxon and sampled-global scales. We then asked whether those decomposed traits can be summarized by one whole-capitulum syndrome or retain partly independent spatial structure.

Location

Global.

Time period

Observations dated 1985–2026; climate normals for 1981–2010.

Major taxa studied

Source-assigned taxa of *Cirsium* and allied Cardueae, used as a replicated-radiation model system rather than as a genus-specific endpoint.

Methods

An object detector localized capitula and deterministic image functions measured a frozen 27-endpoint continuous-trait contract spanning orientation, visible colour, outline and finer involucral and armature-like image geometry. The final trait universe contained 46,276 spatially thinned observations from 259 taxa; 22 endpoints were measured and five remained explicit unexecuted rows. We quantified observation-level variation compressed by taxon means and retained coordinates to map endpoint support. Within- and among-taxon models tested nine predictors in six predeclared blocks: thermal, hydric, radiative/atmospheric, mechanical, growing-season water input and resource/productivity. Global multiplicity correction was followed by broad-space and historical-placement gates. Secondary complete-18 analyses evaluated taxon morphospace, within/among measurement-module organization and environmental redundancy.

Results

Across the 22 measured endpoints, 0.814–0.986 of raw observation-level variation occurred below taxon means; equal-replication endpoint medians were 0.287–0.585. Across the canonical 234-row endpoint-gradient grid, seven pairs were supported at both scales, 18 within taxa only, three among taxa only, 152 at neither scale and 54 were not comparable. Sampling was uneven: Europe plus North America contributed 92.26% of observations and the two most observed taxa 54.03%. Across 674 sampling perturbations, 16/26 selected within-taxon and 10/10 selected among-taxon pairs retained direction throughout. Five within-taxon pairs passed the broad-space screen, but only four also retained direction in every sampling scenario. Two among-taxon

pairs—lower chroma with higher shortwave radiation and a larger signed head-axis angle relative to EXIF-image vertical with higher annual precipitation—passed the broad-space screen, retained direction across sampling perturbations and retained direction and $P < 0.05$ on all 52 historical-placement trees. The expanded 127-taxon $\times$ 18-endpoint morphospace remained multidimensional: its first three principal components explained 42.3% cumulatively, and within- and among-taxon association geometries were only partly aligned (Spearman = 0.3663).

Main conclusions

Global public photographs can move broad-scale trait ecology beyond species means and coarse categories by preserving repeated quantitative distributions and locating each continuous phenotype in present geographic and environmental space. Environmental organization is trait- and scale-dependent, not one climate-associated capitulum syndrome; whole-capitulum analyses provide a secondary synthesis rather than replacing decomposed traits. In a group containing geographically replicated rapid regional radiations, this decomposition identifies the phenotype axes that can later be tested for repeated diversification across distinct lineages without assuming that present spatial associations already demonstrate adaptive radiation. The two final rows are adaptive-pattern candidates under current sequential controls, not demonstrations of adaptation or mechanism. Future work should validate the candidate functions of these continuous components, reconstruct their histories using nuclear ancestry data, distinguish independent origins from retention or reticulation, and reserve convergence or adaptive-convergence claims for analyses with comparable selection pressures and reproductive-fitness evidence.

1 Introduction

Comparative trait ecology links organismal attributes to environmental filtering, species interactions and ecosystem functioning, but broad syntheses often represent each species by one or a few characteristic values. This simplification is useful when measurements are sparse, yet it can conceal variation among populations and individuals that affects niche breadth, interaction strengths and coexistence (Messier et al. 2010; Albert et al. 2011; Bolnick et al. 2011; Violle et al. 2012). Intraspecific variation can constitute a substantial share of community-level plant trait variation (Siefert et al. 2015). Global resources such as TRY and BIEN have transformed cross-species ecology (Kattge et al. 2011, 2020; Maitner et al. 2018), but their broad-scale utility necessarily relies heavily on taxon-level summaries and unevenly replicated measurements. A complementary challenge is therefore to recover the geographic distribution of phenotype itself rather than only a representative value for each species.

A second simplification is categorical description. Many conspicuous plant characters are recorded as states—upright versus nodding, pale versus pigmented, globose versus elongate, spiny versus weakly armed—even when the underlying phenotype varies continuously. Categories are valuable for taxonomy and identification, but they discard distances within and between states and can obscure spatial variation within a named taxon. Moving from categories to

explicitly defined numerical traits makes it possible to ask not only which forms occur, but how much phenotype varies, where that variation occurs and whether different components of morphology occupy different environmental spaces.

Public biodiversity photographs provide an unusually large source of repeated observations for this purpose. Computer-vision studies have demonstrated detection and taxonomic recognition in heterogeneous citizen-science imagery (Van Horn et al. 2018; Wäldchen & Mäder 2018; Rzanny et al. 2019), and multimodal methods are beginning to infer continuous plant traits (Sharma et al. 2026). If each assessable structure can instead be converted into a numerical phenotype—an angle, colour coordinate, outline statistic or contour metric—photographs can become a distributed phenotypic sampling network. Their value is then not simply the number of species represented, but the ability to retain repeated measurements across populations, regions and environmental gradients and thereby construct a quantitative geography of visible phenotype.

This opportunity is inseparable from sampling and measurement bias. Citizen-science records reflect observer behaviour, accessibility and taxonomic preference as well as organism occurrence (Troudet et al. 2017; Di Cecco et al. 2021). Illumination, camera processing, occlusion and viewpoint can alter image-derived colour, orientation and contour measurements. An unassessable photograph is not evidence that a trait is biologically absent. A defensible image-phenomics workflow must therefore preserve trait-specific missingness, measurement provenance and the nesting of heads within photographs, and must interpret outputs as visible image phenotypes rather than error-free biological measurements. These limitations constrain biological interpretation, but they do not remove the descriptive value of measuring phenotype continuously when uncertainty and assessability are explicit.

Trait–environment associations must also be separated by biological scale. A relationship can arise because observations assigned to the same taxon vary across space, because taxa with different median phenotypes occupy different environments, or because both traits and environments reflect shared history. These are complementary descriptions of phenotypic geography, not interchangeable causal tests. Within-taxon analyses isolate spatial covariation after removing time-invariant taxon differences, whereas among-taxon analyses describe environmental sorting of taxon-level phenotypes and require additional attention to phylogenetic non-independence and uncertain placement. A within-taxon gradient is not automatically the small-scale version of taxon turnover, and an among-taxon association cannot be read backwards as plasticity.

A further challenge arises when several measurements belong to one complex reproductive structure. The Asteraceae capitulum is a flower-like pseudanthium in which orientation, visible corolla traits, gross outline and involucre architecture are expressed together. Continuous decomposition is valuable precisely because those components need not share the same geography or environmental gradient. A final whole-capitulum synthesis can test whether the endpoints are partly organized, but it must not erase the trait-specific patterns by forcing them into one syndrome. Floral integration theory motivates partial organization among subsets of

traits (Diggle 2014); whether that organization is similar within and among taxa remains a secondary empirical question rather than the starting unit of inference.

We use thistles as a comparative model system rather than because the biological objective is to explain one genus. The *Carduus*–*Cirsium* group combines a conspicuous, repeatedly photographed capitulum with unusually large visible disparity in orientation, colour, outline and involucre architecture, so multiple phenotype dimensions can be measured on the same reproductive structure. More importantly, its diversity includes geographically replicated rapid radiations rather than one single lineage-specific diversification event. Genus-wide phylogenomics identifies separate rapid Pleistocene radiations after dispersal into Japan and North America, with geographic isolation and ecological specialization implicated in their diversification (Moreyra et al. 2025). North American *Cirsium* has independently been described as a rapid continent-wide radiation in which precipitation and semiaridity were associated with diversification dynamics (Siniscalchi et al. 2023). The Japanese radiation likewise followed an early-Pleistocene colonization and generated extensive regional endemism over a short evolutionary interval. These results do not by themselves establish that every regional radiation is an adaptive radiation. They do, however, make thistles a strong stress test for a broader question: whether repeated phenotypic diversification across distinct regional lineages is compressed when complex morphology is reduced to categorical states or one taxon mean, and whether its present environmental geography is repeated or scale-specific.

The same model system is useful because its component traits have plausible but non-identical ecological roles. Floral orientation can modify radiation load, temperature, precipitation exposure, pollinator presentation and reproductive performance (Chen et al. 2013; van der Kooi et al. 2019; Creux et al. 2021); summer-blooming *Cardueae*, including *Cirsium*, can maintain capitula below surrounding air temperature under hot conditions (Herrera 2025). Visible floral colour may integrate temperature, drought, soil chemistry, herbivory and pollinator responses (Narbona et al. 2018; Vaidya et al. 2018; Borghi et al. 2019; Dalrymple et al. 2020; Sullivan & Koski 2021; Grossenbacher et al. 2026). In *Cirsium palustre*, purple, white and intermediate morphs varied with exposure, population size and year, and white flowers received preferential pollination in some local conditions (Mogford 1972, 1974a,b, 1978). White-flowered forms are also documented in East Asian *Cirsium* (Kitamura 1932), but colour alone does not delimit evolutionary lineages in the Taiwanese *C. japonicum* complex (Chang et al. 2026). Floral displays and scent can attract both pollinators and antagonists (Ohashi & Yahara 1998, 2000, 2002; Theis 2006; Theis et al. 2007), while visually defensive structures need not have their assumed function (Hanley et al. 2007). These studies motivate candidate functional interpretations without determining in advance which phenotype dimensions should covary globally.

Environmental representation presents a related problem. A small set of bioclimatic variables provides a transparent low-dimensional baseline, whereas adding many correlated rasters can turn a biological analysis into a post hoc screen. Mean and seasonal temperature and precipitation

summarize broad climate, but radiation, atmospheric drying, wind, growing-season water input and potential productivity may carry additional process-relevant information. The appropriate test is therefore not whether one added predictor happens to be significant, but whether a predeclared process representation adds multivariate information beyond the low-dimensional core, and whether that redundancy result differs within and among taxa.

Rapid diversification also makes historical control important without making history the endpoint of this study. New World *Cirsium* exhibits high ecological diversity despite low nuclear-rDNA divergence (Kelch & Baldwin 2003), while hybridization, polyploidy and incomplete sampling complicate a single bifurcating account of trait history (Ackerfield et al. 2020; Bureš et al. 2024; Chang et al. 2026). We therefore use available phylogenetic placements as a robustness gate for present among-taxon associations rather than as evidence that the observed spatial patterns identify repeated origins, adaptive radiation or convergence. Those historical questions require denser nuclear ancestry and explicit origin-class tests downstream.

We therefore developed a global public-image phenomics workflow whose first goal is descriptive but quantitatively explicit: to replace coarse state assignments and single taxon means with repeated continuous measurements of visible capitulum phenotype. Object detection localizes visible capitula, deterministic functions quantify orientation, visible corolla colour and outline, and suitable images provide additional continuous measurements of involucre and armature geometry when assessable. The resulting framework treats each trait as a defined numerical variable with its own image requirements and failure state rather than forcing all photographs into common categories.

We asked four linked questions. First, how much repeated visible phenotype diversity is lost when thistle taxa are represented by categories or one taxon mean? Second, which predeclared environmental gradients align with orientation, visible colour, outline and finer involucre or armature-like image geometry, including unsupported trait–gradient pairs? Third, at which level—head, photograph, within-taxon geography, among-taxon turnover or the sampled global domain—is each pattern expressed, and how robust is it to spatial, regional and incomplete historical controls? Fourth, after those components have been resolved, can they be summarized by one whole-capitulum syndrome, or is the phenotype multidimensional and only partially organized across scales?

These questions define the endpoint of the present study: a quantitative reconstruction of the current spatial state of continuous capitulum traits in which each trait definition, uncertainty, distribution, environmental association, scale class and robustness status remains explicit. Future work can use these bounded measurements to validate candidate functions and reconstruct evolutionary histories, but claims about independent origins, convergence or adaptive convergence require nuclear ancestry, explicit origin-class tests, comparable selection pressures and reproductive-fitness evidence. Neither spatial association nor repeated history alone establishes cause.

2 Methods

2.1 Study design and spatial cohort

We used public biodiversity photographs to quantify continuous capitulum phenotypes across source-assigned taxa of *Cirsium* and allied Cardueae. Source names were retained as operational taxonomic units because the photograph records do not resolve a genus-wide taxonomy. The analysis unit was a coordinate-bearing observation rather than one species-level database row. The exhaustive source stream contained 406,582 detector-positive observations from 286 taxa. Coordinate filtering retained 392,989 observations from 271 taxa, and a positional-accuracy threshold of 10 km retained 297,293 observations from 259 taxa. Before examining trait values, we retained one observation per taxon by 0.25-degree cell, yielding the frozen spatial cohort of 46,276 observations from 259 taxa. Dates ranged from 1985 to 2026. Missing or unassessable image measurements were never encoded as biological absence or zero (Figure 2; Table 1; Table S2).

2.2 From photographs to a 27-endpoint continuous-trait contract

A single-class YOLO11n detector localized visible capitula at a confidence threshold of 0.25. Detection defined image regions of interest only; downstream traits were calculated with deterministic image functions. The category-free contract registered 27 continuous endpoints across orientation, visible corolla colour, gross outline, floral display, involucre architecture, armature-like projection geometry and validation-only surface-image modules. Orientation was the signed head-axis angle relative to EXIF-oriented image vertical: 0 degrees upward, 90 degrees horizontal and 180 degrees downward. Image vertical is reproducible but is not a gravity measurement. Colour was represented in visible CIELAB space by lightness, chroma and the sine and cosine of circular hue. Outline and contour variables were dimensionless ratios or normalized geometry. Fine involucre, armature-like and surface endpoints were retained as image phenotypes and were not relabelled as botanical spine length, stiffness, hair density, glands or secretion.

The final artifact-backed run measured 22 of 27 endpoints. Eighteen of 19 inferential endpoints were available; `visible_floret_fraction` and four descriptive colour-composition fractions remained explicit `unexecuted_no_measurement` rows. Hue sine and cosine formed one joint circular inferential unit, producing 26 units from the 27 registered endpoints (Figure 1; Table S1).

2.3 Trait geography and information retained below taxon means

For every endpoint we recorded observation and taxon counts, latitude and longitude limits, and occupancy in 5-degree geographic cells. To quantify information compressed by a one-row-per-taxon database, we decomposed total observation-level sums of squares into variation below and between source-assigned taxon means. Because abundant taxa could dominate the raw partition, we also ran 500 deterministic equal-replication resamples with two observations per eligible

taxon. These quantities describe visible image-phenotype variation; they are not heritability, genetic variance or plasticity estimates (Figure 3; Table S3).

2.4 Six predeclared environmental blocks

Nine predictors formed six biological hypothesis blocks. Thermal conditions comprised annual mean temperature and temperature seasonality (CHELSA BIO1 and BIO4). Hydric conditions comprised annual precipitation and precipitation seasonality (BIO12 and BIO15). Radiative and atmospheric exposure comprised mean shortwave radiation and vapour-pressure deficit. Mechanical exposure was represented by mean near-surface wind. Growing-season water input was represented by climatic growing-season precipitation, and resource/productivity by modelled potential net primary productivity. Growing-season precipitation was not taxon-specific flowering-season rainfall, and potential NPP was not measured local resource supply.

All predictors had to retain at least 98% coverage in the phenotype-blind spatial cohort. The nine-variable environmental table was constructed before endpoint-specific missingness filtering; no unavailable variable could be substituted after observing trait associations.

2.5 Within- and among-taxon environmental associations

Within-taxon models used taxon-demeaned standardized traits and predictors, no intercept, and taxon-clustered standard errors. Eligibility required at least 100 observations, ten taxa and two observations per retained taxon. Joint hue used the magnitude and direction of its standardized sine and cosine coefficients under 9,999 permutations restricted within taxa.

Among-taxon models paired taxon trait medians with taxon environmental medians. The primary scope required at least five trait observations per taxon and at least 20 taxa; a separately corrected sensitivity required two observations. Linear endpoints used standardized univariate slopes and 9,999 taxon-label permutations. Joint hue used paired permutations of the sine and cosine slopes. Benjamini-Hochberg correction formed one global family across all successful unit by predictor rows separately for within taxa, among-taxon minimum-five and among-taxon minimum-two scopes. Historical endpoint tiers did not split multiplicity or remove rows. Each primary minimum-five row was classified as supported at both scales, within taxa only, among taxa only, neither, or not comparable (Figure 4; Tables S4–S6).

2.6 Sampling-composition sensitivity

Every globally FDR-supported row entered a separately frozen retrospective sensitivity; all atlas rows remained reported regardless of entry. Sampling perturbations were selected without using trait outcomes. Within-taxon rows were refitted with equal total weight per taxon. Both scales omitted each of the ten most observed taxa separately and the two most observed taxa jointly, omitted each of six broad geographic regions separately, and retained only observations explicitly classified as native by the frozen WCVF/TDWG join. Linear rows had to retain coefficient sign; circular rows had to remain within 90 degrees of the original coefficient vector. We reported

direction and effect-magnitude stability only and calculated no post-selection P or q values (Figure 5; Table S7; Figure S2).

2.7 Broad-space and historical-placement sensitivity

Globally FDR-supported rows were refitted with a second-order spherical-coordinate basis. Passage required a spatial permutation P below 0.05, preserved sign for a linear endpoint and residual eight-nearest-neighbour Moran P of at least 0.05. This was a broad geographic sensitivity, not removal of every spatial process.

Among-taxon rows passing the broad-space gate were then fitted by univariate Pagel-lambda PGLS on 52 audited dated-backbone placement trees: two deterministic scenarios and 50 randomized within-genus placements. A final candidate had to retain direction and P below 0.05 on every tree. Because only 54 historical-atlas taxa were direct backbone tips, this procedure is a placement sensitivity rather than a resolved species-tree correction (Figure 5; Table 2; Table S8; Figures S3-S4).

2.8 Secondary whole-capitulum synthesis

After endpoint-level inference was complete, a secondary complete-18 analysis asked whether the measured traits reduced to one capitulum syndrome. Standardized taxon medians were analysed by principal components, and within- versus among-taxon association matrices were compared by measurement module. This synthesis did not select endpoints or alter the endpoint-level multiplicity families (Table S9).

2.9 Claim boundary

The study is a retrospective exploratory analysis of visible image phenotypes and their present spatial associations. It does not identify plasticity, local adaptation, selection, causal environmental mechanisms, botanical identity of validation-only proxies, independent evolutionary origins or convergence. The two final rows are termed adaptive-pattern candidates under current sequential controls, not demonstrations of adaptation.

3 Results

3.1 Continuous photographs recovered variation hidden by taxon means

The frozen spatial cohort contained 46,276 observations from 259 source-assigned taxa. Twenty-two of 27 registered endpoints were measured, producing 374,255 analysis-eligible observation by endpoint values. Coverage ranged from 909 observations and 85 taxa for validation-only surface proxies to 40,510 observations and 253 taxa for visible CIELAB lightness and chroma. Measured endpoints occupied 190 to 431 five-degree cells (Figures 1–2; Tables S1–S2; Figure S1).

Across the 22 measured endpoints, 0.8137 to 0.9863 of raw observation-level variation occurred below source-assigned taxon means. The equal-replication median ranged from 0.2869 for hue sine to 0.5846 for basal taper ratio. Thus, even after limiting each eligible taxon to two observations, one taxon mean compressed 28.7–58.5% of visible variation, while the unbalanced observed sample compressed 81.4–98.6% (Figure 3; Table S3).

3.2 Environmental associations were endpoint- and scale-specific

Eight predictors had complete coverage and wind had 99.989% coverage. The within-taxon atlas contained 189 executed tests and 45 explicit unexecuted rows; 26 executed rows passed the global FDR family. Supported rows occurred in all six blocks, including orientation with annual mean temperature, chroma with VPD, several involucre geometry endpoints with precipitation metrics, surface-image endpoints with temperature or growing-season precipitation, and joint hue across multiple gradients.

The primary among-taxon atlas contained 180 successful tests, 45 unexecuted rows and nine non-finite-variation rows. Ten rows passed its separate global FDR family. Linear associations comprised a larger orientation angle with annual precipitation (standardized effect 0.3044, $q = 0.0210$), a larger orientation angle with growing-season precipitation (0.2581, $q = 0.0338$), lower chroma with higher shortwave radiation (-0.3454 , $q = 0.0060$), and greater bract projection-peak density with VPD (0.4437, $q = 0.0468$). Joint hue was associated with BIO1, BIO4, BIO15, radiation, wind and NPP. Seven rows repeated in the minimum-two sensitivity.

Across the complete 234-row primary scale grid, seven rows were supported at both scales, 18 within taxa only, three among taxa only, 152 at neither scale and 54 were not comparable. The three among-only rows were orientation with annual precipitation, orientation with growing-season precipitation and chroma with radiation. Most supported involucre and surface-proxy rows were within-taxon only (Figure 4; Tables S4–S6).

3.3 Sampling was extensive but strongly uneven

Europe contributed 23,337 observations (50.43%) and North America 19,357 (41.83%); together they contributed 92.26%. The Northern Hemisphere contributed 94.91%. *Cirsium vulgare* contributed 28.54%, *C. arvense* 25.49% and the ten most observed taxa 79.93%. The taxon-level median was six observations; 51 of 259 taxa had one observation and 150 had fewer than ten.

All 674 declared sampling scenarios were evaluable. All ten selected among-taxon pairs retained direction in every applicable scenario. Sixteen of 26 selected within-taxon pairs retained direction throughout; ten reversed in at least one scenario. Four of the five within-taxon broad-space passes were also directionally stable throughout. Projection roughness with radiation reversed under equal taxon weighting and after jointly omitting the two dominant taxa (Table S7; Figure S2).

3.4 Broad-space and historical gates retained two among-taxon candidates

Five within-taxon rows passed the broad-space gate: orientation-BIO1, chroma-VPD, involucre length/width-BIO15, projection roughness-radiation and surface high-frequency energy-BIO1.

Their spatial standardized effects were 0.0265, -0.0421, 0.0602, -0.1392 and 0.1235, respectively. These rows describe within-taxon spatial covariation and were not treated as plasticity or local adaptation.

Two among-taxon rows passed the broad-space gate. Chroma decreased with radiation (spatial $\beta = -0.7124$, spatial $P = 0.001$, residual Moran $P = 0.217$), and orientation angle increased with annual precipitation (spatial $\beta = 0.2861$, spatial $P = 0.017$, residual Moran $P = 0.082$). Both retained direction in all 18 applicable sampling scenarios; their minimum effect-magnitude ratios were 0.615 and 0.781.

Both rows retained direction and P below 0.05 on all 52 historical-placement trees. Chroma-radiation PGLS β was -0.3454 with $P = 0.000024$ and Pagel $\lambda = 0$ throughout. Orientation-precipitation PGLS β ranged from 0.3044 to 0.3045, P from 0.000201 to 0.000231 and λ from 0 to 0.0537. They are therefore adaptive-pattern candidates under the current sequential controls, not evidence that an adaptive mechanism or independent origin has been demonstrated (Figure 5; Table 2; Table S8; Figure S4).

3.5 Whole-capitulum synthesis did not collapse the trait atlas

The expanded 127-taxon by 18-endpoint morphospace remained multidimensional: the first three principal components explained 42.3% cumulatively. Measurement-module contrasts were modest, and the correspondence between within- and among-taxon association geometry was partial (Spearman = 0.3663). Whole-capitulum structure therefore summarized, but did not replace, the decomposed endpoint results (Table S9).

4 Discussion

4.1 A method for spatially explicit continuous phenomics

The primary contribution is a change in what broad-scale trait data preserve. Instead of assigning one categorical state or mean to each taxon, public photographs yielded repeated numerical measurements with coordinates and endpoint-specific missingness. The raw and equal-replication partitions show that substantial visible phenotype information sits below taxon means. This information is unavailable to a species-mean database even when the database value itself is accurate.

The method is not merely an automated scoring device. It connects a declared image measurement, its assessability and uncertainty, repeated within-taxon distributions, mapped geographic support and an explicit within- versus among-taxon estimand. That connection makes it possible to separate a present within-taxon gradient from taxon turnover and to display unsupported rows without converting missing observations to trait absence.

4.2 There is no single climate-associated capitulum syndrome

The 234-row scale grid rejects a scale-invariant whole-head response. Most supported involucral and surface-image associations occurred only within taxa, while orientation-precipitation, orientation-growing-season precipitation and chroma-radiation occurred only among taxa. Seven rows were supported at both scales, but most involved joint visible hue and several retained residual spatial structure. A coefficient at one biological scale therefore cannot be read automatically as the process operating at another.

The secondary 18-endpoint synthesis reinforces this conclusion. The first three principal components captured less than half of taxon-level variation, and within- and among-taxon association geometries were only partly aligned. A whole-capitulum summary is useful for describing partial organization, but it should not erase the endpoint-specific geography that made the result visible.

This multidimensionality matters especially because thistles contain geographically replicated rapid regional radiations rather than one isolated diversification event. Separate radiations in Japan and North America create a downstream opportunity to ask whether the same phenotype axes repeatedly diversify, whether different radiations explore different parts of capitulum trait space, and whether similar environments recruit the same or different trait combinations. The present analysis does not test those historical questions and therefore does not infer adaptive radiation from current geography. Its contribution is to define the continuous phenotype axes and scale-specific environmental patterns that such repeated-radiation tests would otherwise have to reconstruct from coarse taxonomic descriptions.

4.3 The two candidates are robust patterns, not demonstrated mechanisms

Taxa from higher-radiation environments had lower photographed CIELAB corolla chroma. Radiation-responsive floral pigment regulation is biologically plausible: UV-B can induce petal anthocyanin accumulation, and geographic variation in floral anthocyanins can covary with solar UV (Hennayake et al. 2006; Peach et al. 2020). However, visible chroma is a colour-space distance, not an anthocyanin assay or UV-reflectance measurement. The negative coefficient cannot be translated directly into more or less pigment synthesis. Calibrated visible and UV reflectance, petal chemistry or pathway expression, and common-garden radiation manipulation are required.

Taxa from wetter environments had a larger angle on the frozen 0-up, 90-horizontal, 180-down image scale, which is consistent with more downward presentation. Experimental work in a nodding Asteraceae supports rain shielding as a plausible function of downward flowers (Chen et al. 2013). Yet EXIF image vertical is not gravity, and the present study did not measure flowering-period rain interception, pollen wetting, viability or fitness. Gravity-referenced three-dimensional orientation and experimental angle manipulation are required before a rain-protection mechanism can be claimed.

Both candidates survived global multiplicity correction, broad-space screening, residual spatial diagnostics, all tested sampling-composition directions and 52 historical-placement trees. Those checks make the patterns difficult to explain by the particular tested sampling, broad spatial trend or within-genus grafting scenario. They do not establish heritability, selection, fitness advantage, causal mediation or independent evolutionary origin.

4.4 Uneven citizen-science sampling remains part of the result

The dataset is geographically extensive but not a probability sample. Europe and North America supplied more than 92% of observations, two taxa supplied 54%, and many taxa were sparsely represented. The retrospective sampling audit makes that imbalance visible rather than claiming to remove it. Directional stability for all ten selected among-taxon rows and 16 of 26 selected within-taxon rows is useful, but it does not make the sample globally representative or create a new confirmatory test.

Measurement limitations are equally important. Camera processing affects visible colour; two-dimensional outline depends on viewpoint; image vertical does not supply gravity; and fine involucre and surface variables require independent botanical calibration. Five registered endpoints were unexecuted, including visible floret fraction. Independent detector boxes, reference measurements, developmental-stage labels, paired flower-background colour controls and repeat-photo remeasurement remain required before submission claims can be strengthened.

4.5 Future directions

The present study stops at defined continuous traits, their uncertainty, current geographic support, environmental alignment, biological scale and robustness status. Future work can first validate candidate functions, then reconstruct when and how often supported functional states arose using nuclear ancestry data and distinguish independent origins from retention, ancestral polymorphism and introgression. Convergence or adaptive-convergence claims should be tested only when independent origins, comparable selection pressures and reproductive-fitness evidence are available; spatial association alone cannot supply those historical or causal conclusions.

4.6 Overall implication

Public photographs can recover a quantitative geography of visible phenotype that species categories and means compress. In thistle capitula, the recovered field is multidimensional and scale dependent. The clearest current among-taxon patterns are lower visible chroma under higher radiation and a larger image-referenced orientation angle under higher annual precipitation. Their value is that they now define precise botanical, physiological and evolutionary tests—including whether distinct rapid regional radiations repeatedly diversify along the same trait axes—not that those mechanisms or histories have already been solved.

5 Conclusion and declarations

Starting from 27 registered continuous endpoints, the study recovered 22 measured, coordinate-bearing capitulum phenotypes across 46,276 public observations and 259 source-assigned taxa. Taxon means compressed 81.4-98.6% of raw visible variation and 28.7-58.5% after equalizing replication. Environmental organization was endpoint- and scale-specific rather than one capitulum syndrome. Two among-taxon rows—lower visible chroma with higher shortwave radiation and a larger image-referenced orientation angle with higher annual precipitation—retained support through the current multiplicity, sampling-composition, broad-space and historical-placement gates.

These are adaptive-pattern candidates under current sequential controls, not demonstrations of adaptation or mechanism. The radiation-responsive pigment and rain-shielding interpretations remain testable hypotheses. Citizen-science sampling is uneven, and independent detector, trait-measurement and botanical validation remain open. No claim is made about plasticity, heritability, selection, causal environmental response, independent origins or convergence.

Declarations

Ethics approval was not required for analysis of publicly available biodiversity observations. Source photographs will not be redistributed beyond their individual licence terms. Author contributions, funding, acknowledgements, conflicts of interest and corresponding-author details must be completed by the authors in the journal submission system and separate title page. The current repository package remains a submission candidate on scientific HOLD until the external validation gates listed in `EXTERNAL_COMPLETION_GATES.md` are closed.

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Data and code availability

A versioned submission-candidate bundle containing the blinded Main and Supporting Information files, the formatted Main Tables 1–2 and Tables S1–S10 workbook, derived CSV tables, model outputs, figure sources and SHA-256 reproducibility manifests accompanies this repository. The public GitHub repository is not an anonymous review link; before journal submission, the exact bundle will be copied without Git history or account identifiers to an anonymized read-only location for editors and reviewers. Source photographs will not be redistributed beyond their individual licence terms. GitHub Actions artifacts are temporary execution records and are not the durable archive. After the independent validation gates, author metadata and repository licence are complete, the exact supporting version will be deposited in a durable repository with a persistent digital object identifier, and this statement will be replaced with the final repository name, DOI and access terms.

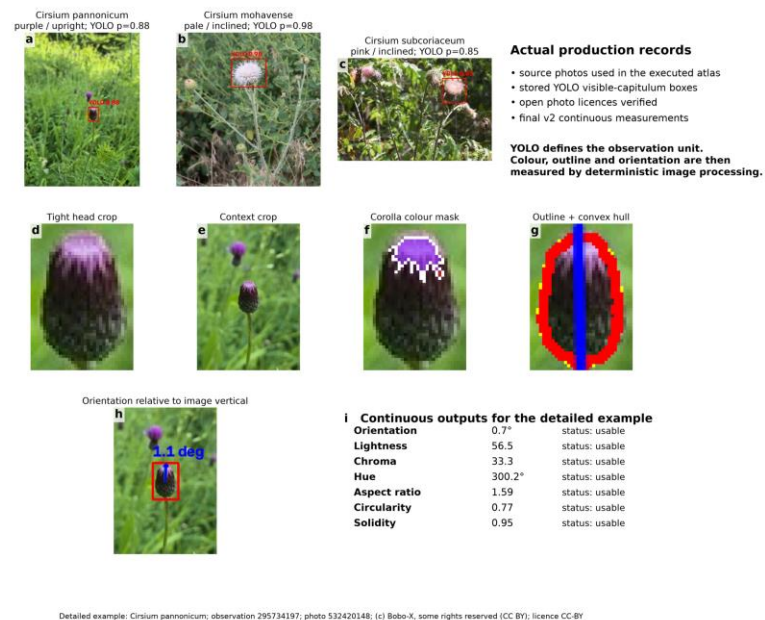
Tables

Table 1. Compact analytical roadmap for the global public-image study of thistle capitula. Each stage identifies its frozen scope, operation or decision gate, and linked display; the full eight-column roadmap is Table S10.

analytical stage	frozen scope	operation and gate	linked display
Photo localization	Visible-capitulum bounding boxes	YOLO localization only; deterministic functions score traits after cropping. Gate: A box defines the image unit; independent detector validation remains an external gate	Figure 1; Tables S1-S2
Cohort and realized domain	46,276 observations; 259 source-assigned taxa	Maximum 10-km positional uncertainty; one observation per taxon by 0.25-degree cell. Gate: Freeze the realized sampling domain before examining endpoint values	Figure 2; Table S2
Continuous measurement contract	27 registered endpoints; 22 measured; five explicit unexecuted	Deterministic orientation, visible-colour, outline and fine-geometry measurements. Gate: Missing or unassessable measurement remains missing evidence	Figure 1; Table S1
Taxon-mean information loss	22 measured endpoints	Raw sums-of-squares partition plus 500 two-observations-per-taxon resamples. Gate: Describe visible image-phenotype information loss, not genetic variance	Figure 3; Table S3
Within/among environmental atlas	26 inferential units by nine predictors in six blocks; 234 canonical rows	Taxon-demeaned clustered models and taxon-label permutations; separate global BH families. Gate: Classify every row as both, within-only, among-only, neither or not-comparable	Figure 4; Tables S4-S6
Sampling-composition audit	674 evaluable retrospective scenarios	Direction and effect-magnitude retention only; no post-selection P values. Gate: Annotate sensitivity without converting the sample into a probability sample	Figure 5; Table S7
Broad-space gate	26 within-taxon and ten among-taxon rows	Spatial-basis refit, permutation P and eight-nearest-neighbour residual Moran diagnostic. Gate: Five within and two among rows pass the declared broad-space screen	Figure 5; Table S8
Historical-placement gate	Two deterministic and 50 randomized within-genus placement trees	Pagel-lambda PGLS sensitivity on 52 trees per row. Gate: Two adaptive-pattern candidates under current controls; not adaptation, mechanism or convergence proof	Figure 5; Table S8

Table 2. Frozen evidence chain for the two among-taxon candidate patterns in the global thistle-capitulum cohort. Beta denotes a standardized coefficient; BH denotes Benjamini-Hochberg; P values are descriptive outputs of the predeclared gates. Stability under these controls is not evidence of adaptation, convergence or mechanism.

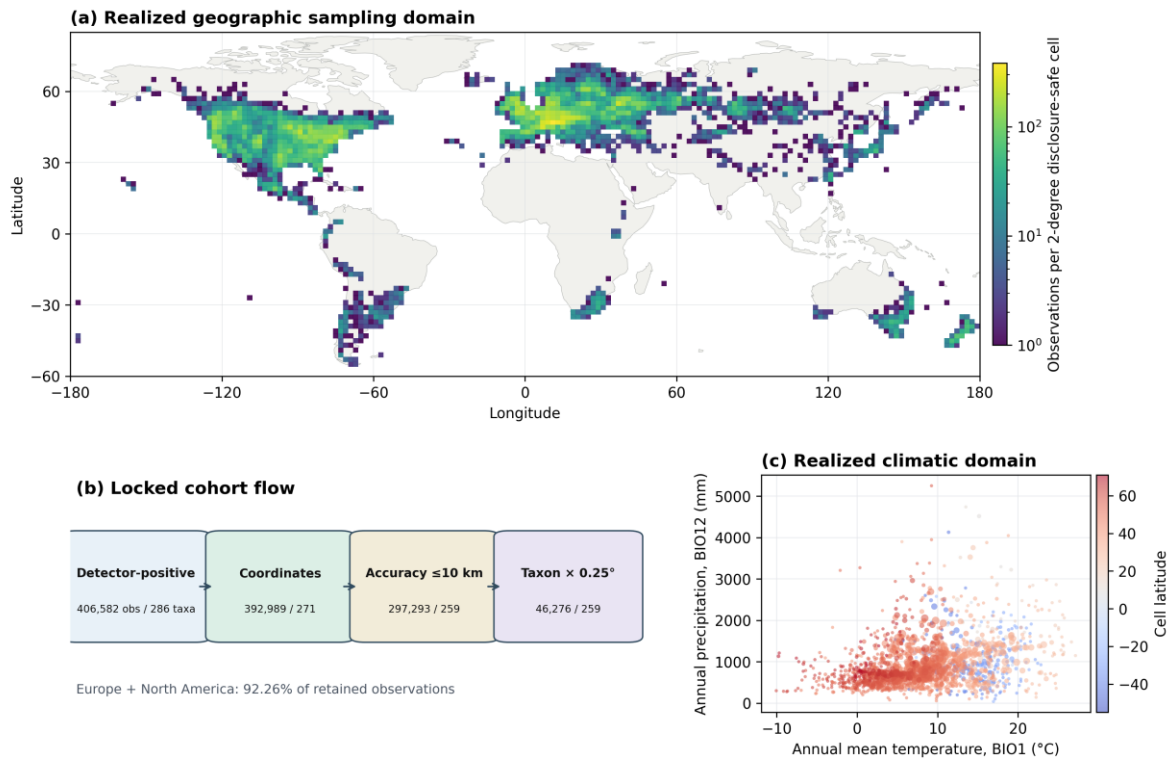
candidate pattern	global among taxon	sampling composition	broad space gate	historical placement
Lower camera-recorded corolla chroma with higher shortwave radiation	n = 143; beta = -0.345; global BH q = 0.006	all declared directions stable; minimum effect-magnitude ratio = 0.615	beta = -0.712; permutation P = 0.001; residual Moran P = 0.217	P < 0.05 on 52/52 trees; P range 2.39e-05-2.39e-05; lambda range 0.000-0.000
Larger image-referenced orientation angle with higher annual precipitation	n = 142; beta = +0.304; global BH q = 0.021	all declared directions stable; minimum effect-magnitude ratio = 0.781	beta = +0.286; permutation P = 0.017; residual Moran P = 0.082	P < 0.05 on 52/52 trees; P range 2.01e-04-2.31e-04; lambda range 0.000-0.054



27 registered continuous endpoints	22 measured finite v2 outputs	5 unexecuted remain missing evidence	Localization only detector does not score traits
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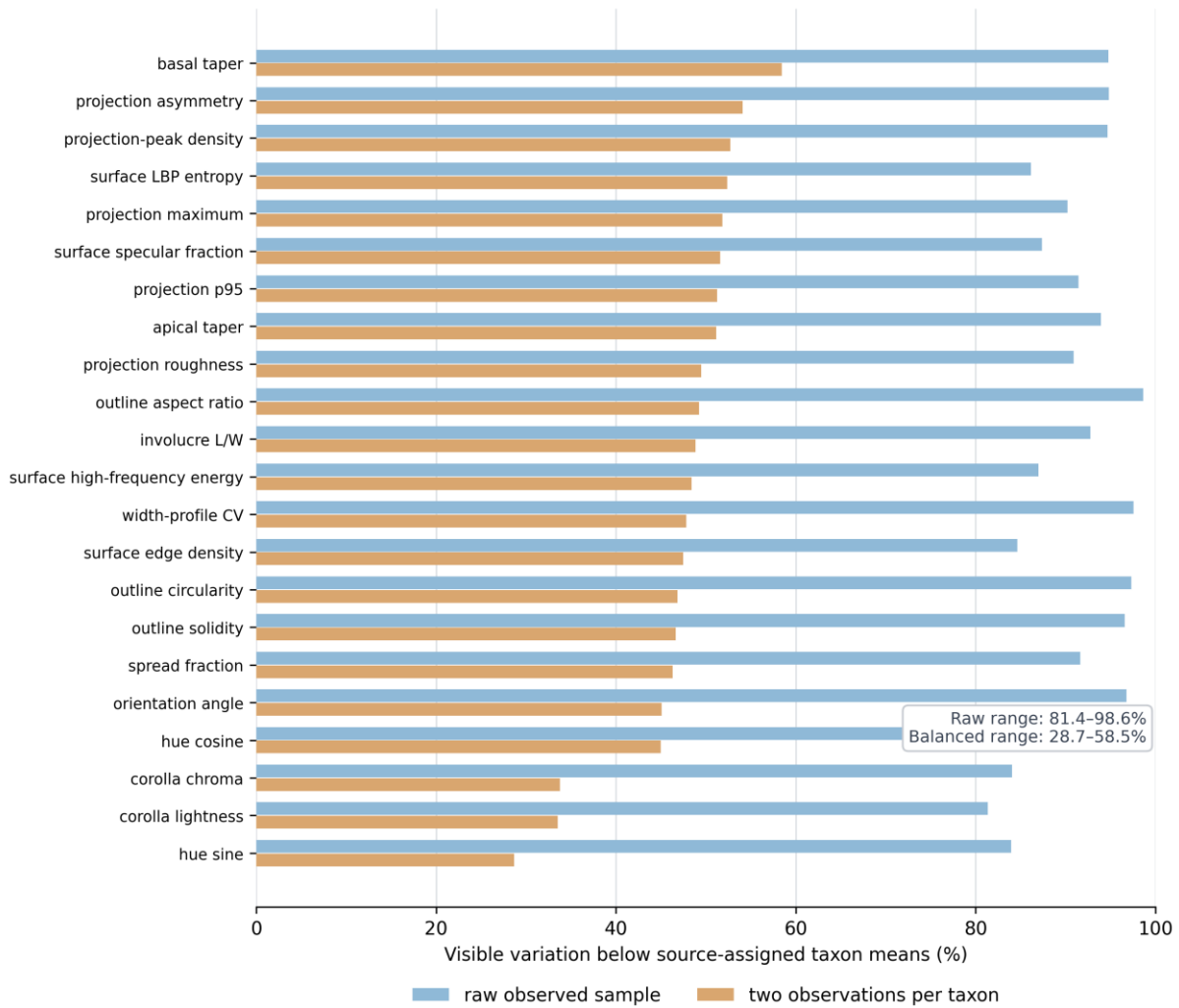
Three open-licensed photographs are presentation/source provenance only; the v2 cohort and results come exclusively from the frozen full-27 lane.

Figure 1. From public photographs to deterministic continuous measurements in the global Carduus-Cirsium thistle-capitulum study. Three open-licensed photographs and their detector/crop examples are presentation/source provenance only; all numerical result statements are current v2.



The panels describe where retained records occur; they do not establish geographic or environmental representativeness.

Figure 2. Realized global sampling domain of the frozen public-image cohort of thistle capitula. (a) Disclosure-safe observation density, (b) locked filtering flow to 46,276 observations and 259 source-assigned taxa, and (c) the realized annual-temperature (BIO1) and annual-precipitation (BIO12) domain. The panels do not establish representativeness.



Observed image-phenotype variation; measurement, photography and biological variation are not separated.

Figure 3. Visible image-phenotype variation compressed by source-assigned taxon means across the global public-image cohort of thistle capitula. Horizontal bars compare the raw observed sample with deterministic equal-replication resamples across 22 measured endpoints; the partition is not genetic variance.

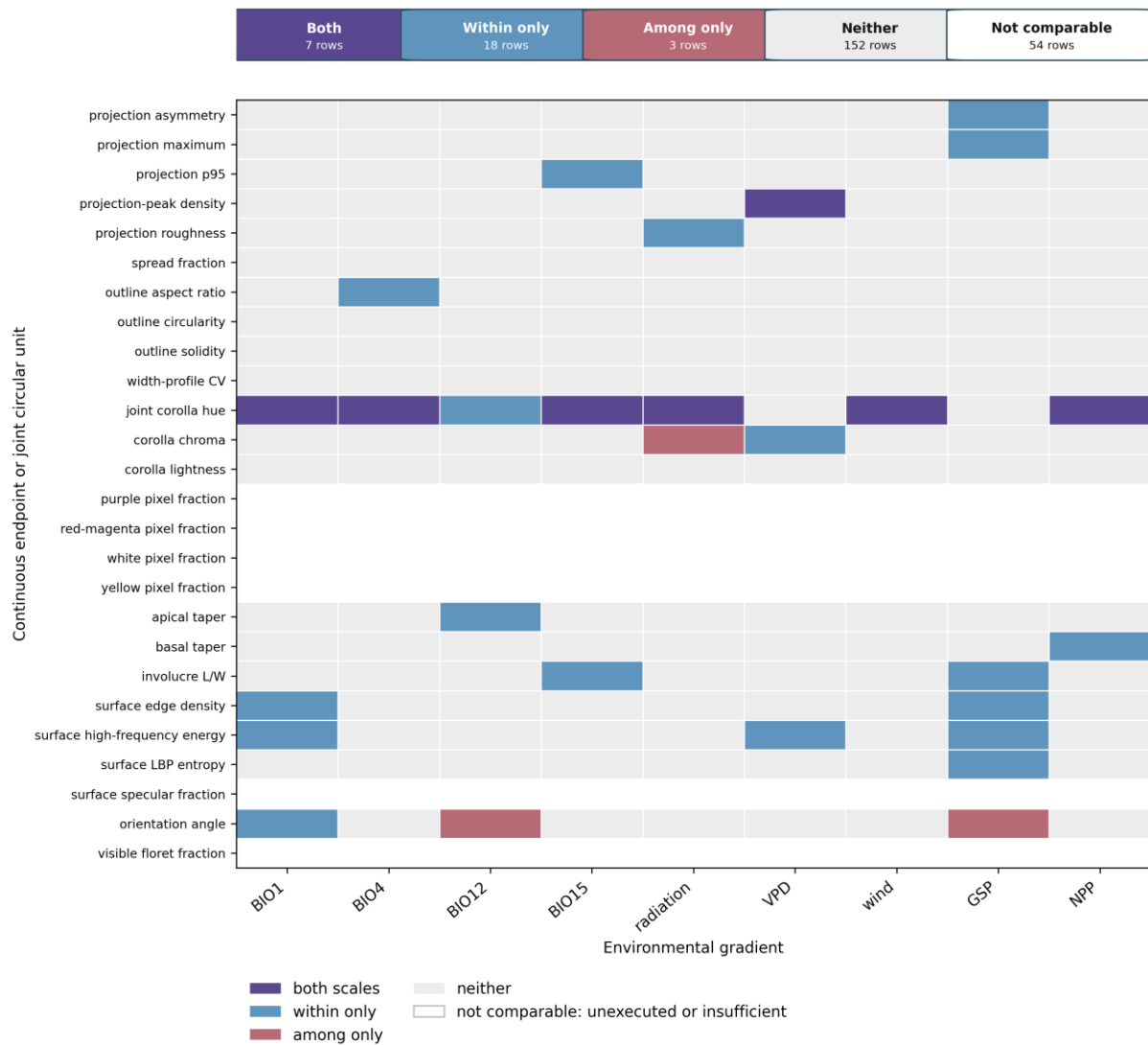


Figure 4. Endpoint-specific environmental alignment within and among source-assigned thistle taxa across the global public-image cohort. The complete 234-row heat map distinguishes both-scale, within-only, among-only, neither and not-comparable rows; joint hue is one circular inferential unit.

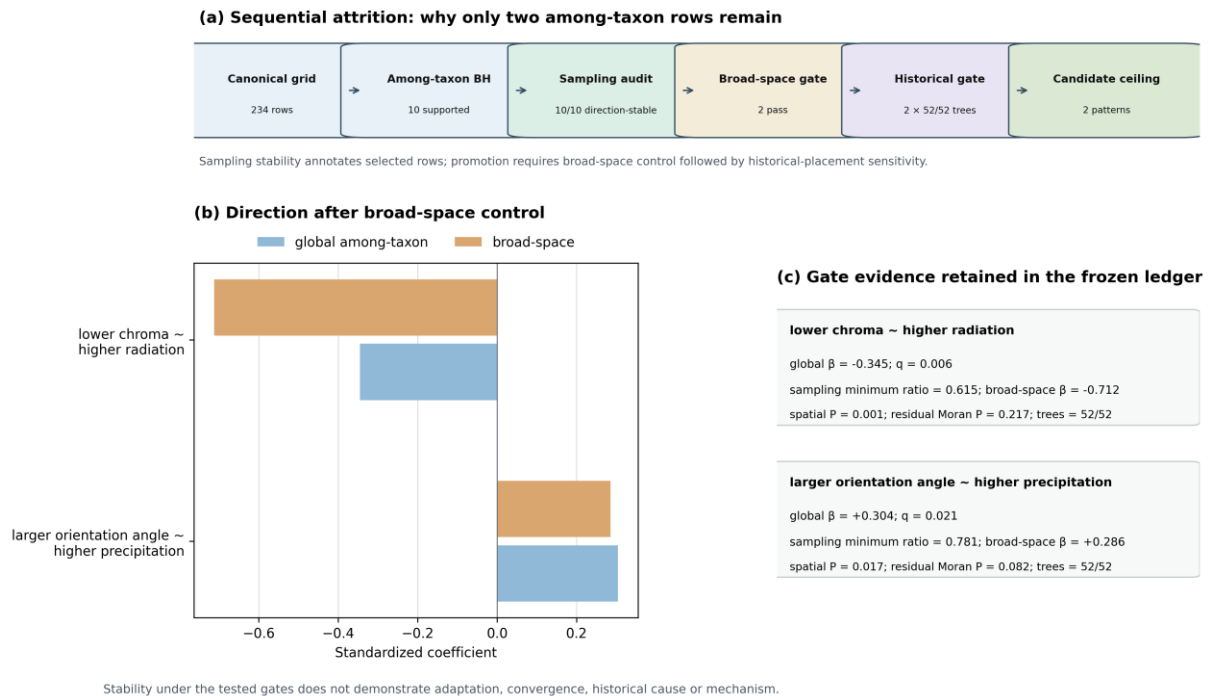


Figure 5. Sequential attrition and robustness of the two final candidate patterns in the global thistle-capitulum cohort. (a) The 234-row grid is reduced through among-taxon multiplicity, sampling annotation, broad-space and 52-tree placement gates; (b-c) retain the coefficient and gate evidence for chroma-radiation and image-referenced orientation-precipitation. Stability is not evidence of adaptation, convergence or mechanism.